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Course	CCSA_104 Security Policy
Date	08-11-2026

Part A: Practical Tasks

Task 1: Explore the default policy

Objective

To examine the existing Access Control policy in SmartConsole, identify the default Cleanup rule, review rulebase columns, and inspect policy management settings.

Policy Package

Setting	Value
Policy Package Name	Standard

Cleanup Rule Details

Field	Value
Name	Cleanup Rule
Source	Any
Destination	Any
Services & Applications	Any
Action	Acce pt
Track	Log
Install On	Policy Target

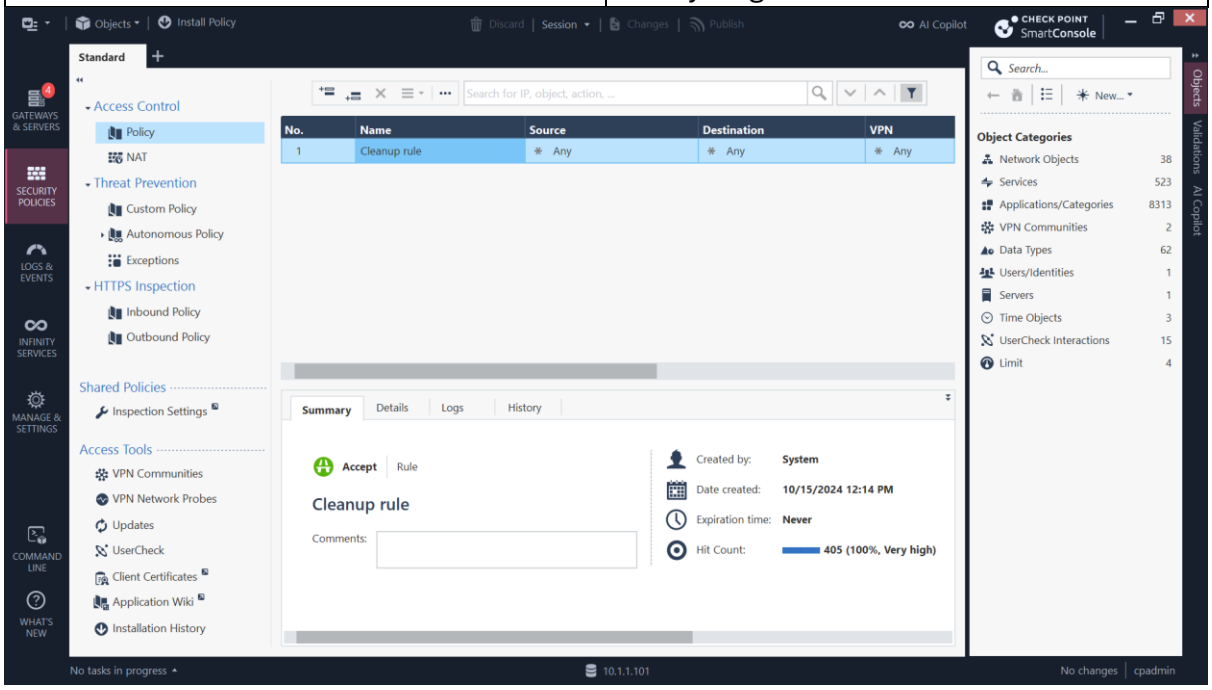
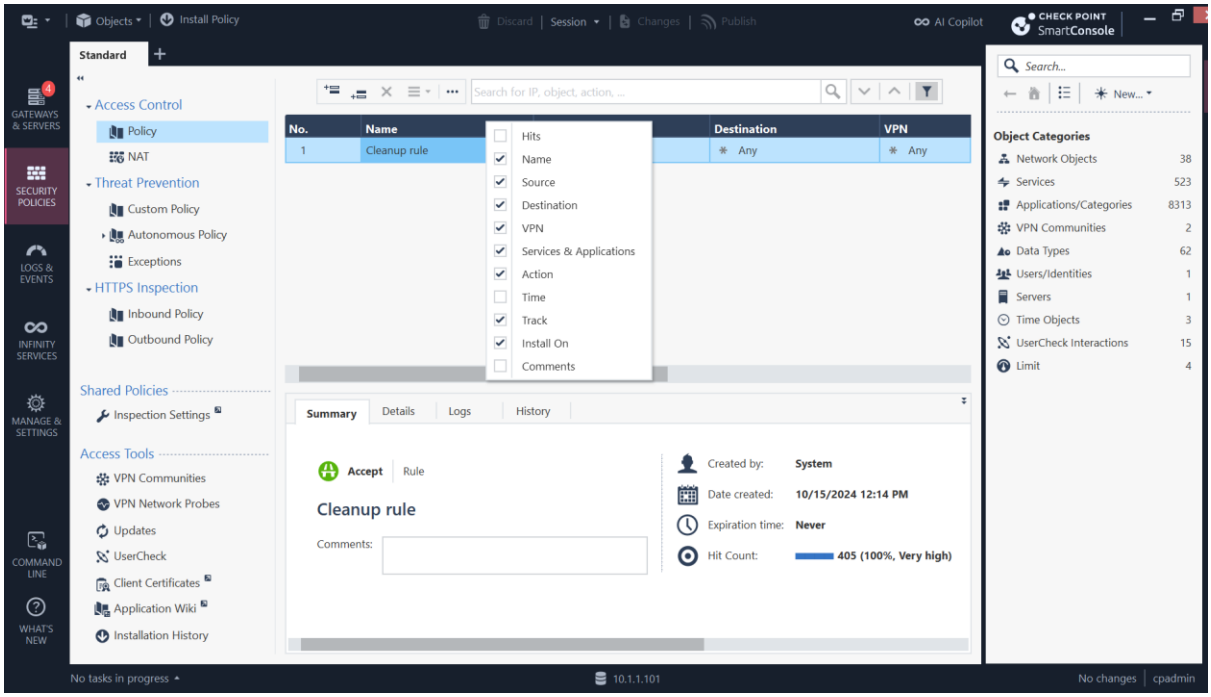


Fig. Default rulebase and Cleanup rules Values

Rulebase Column Review



Global Properties

Three categories observed in Global Properties:

1. Firewall
2. NAT
3. VPN

Policy Packages

Existing policy packages observed:

1. Access Control
2. Threat Prevention
3. QOS

Task 2: Build the Access Control rulebase

Objective

To create and arrange Access Control rules that regulate administrative access, protect the gateway, permit approved business traffic, and log security events.

Positions	Name	Source	Destination	Services & Applications	Action	Track
1	Admin Access	Admin_PC	Gateway	ssh, https	Accept	Log
2	Stealth Rule	Any	Gateway	Any	Drop	Log
3	Inbound to DMZ	Any	Web_Server, Mail_Server	http, https, smtp	Accept	Log
4	Internal to DMZ					
5	DNS Out	Internal_LAN	External_DNS	dns (UDP and TCP)	Accept	Log
6	Internal Web Out	Internal_LAN	Any	http, https	Accept	Log
7	SMTP Out	Mail_Server	Any	smtp	Accept	Log
8	Cleanup Rule	Any	Any	Any	Drop	Log

Procedure

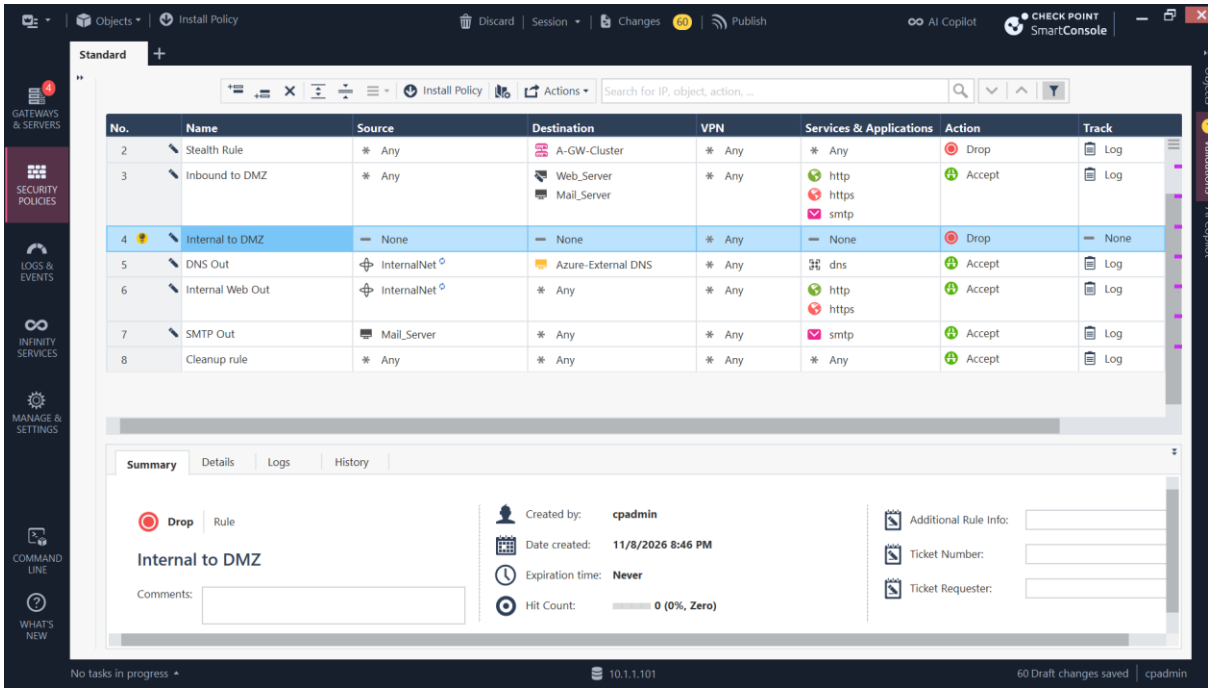
1. Opened Security Policies → Access Control.
2. Added new rules above the existing Cleanup Rule.
3. Configured each rule according to the lab specification.
4. Updated the Cleanup Rule Track setting to Log.
5. Verified the final rule order matched the required positions.

Result

The Access Control rulebase was configured with administrative access controls, gateway protection, DMZ access rules, outbound DNS access, web browsing permissions, and email services. The Cleanup Rule remained at the bottom of the rulebase and was configured to log dropped connections.

Observation

The rulebase follows the top-down evaluation process used by Check Point Security Gateways. Specific access rules are evaluated before the Cleanup Rule, ensuring that legitimate traffic is allowed while unmatched traffic is denied and logged for auditing purposes.



Full rulebase showing:

- Rule numbers
- Rule names
- Source column
- Destination column
- Services & Applications column
- Action column
- Track column

Task 4: Organise the Rulebase with Sections

Objective

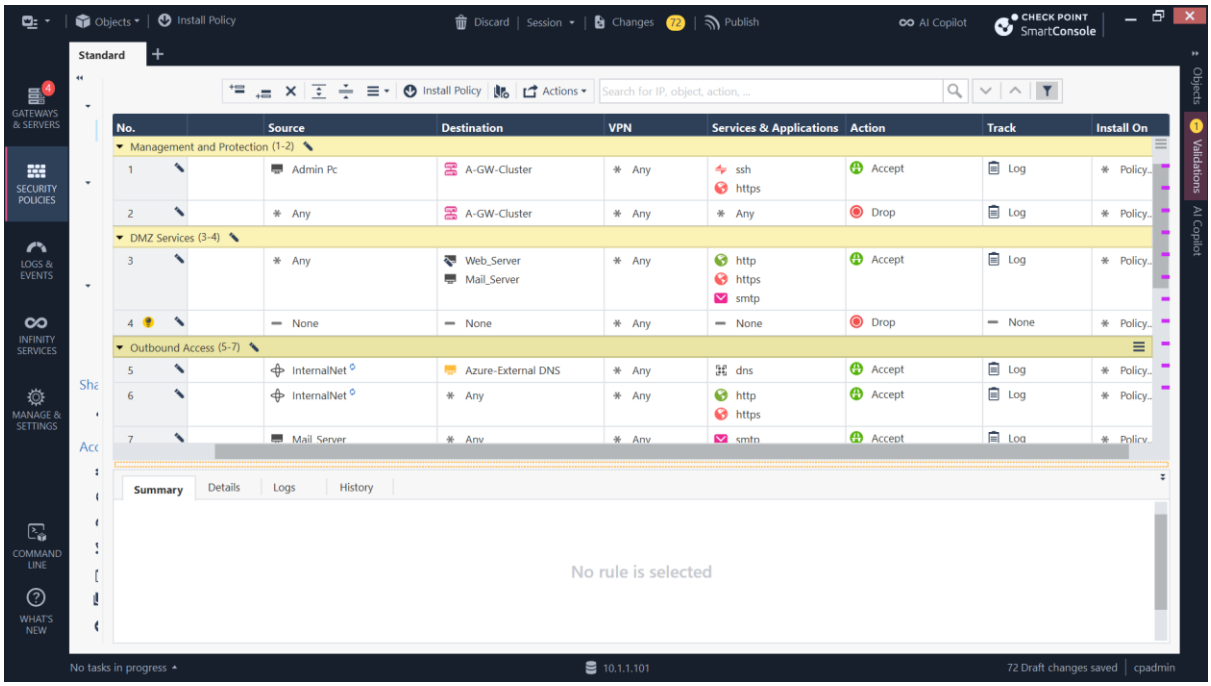
To improve rulebase organization by grouping related security rules into logical sections for easier administration and policy review.

Section Configuration

Section Title	Contains
Management and Protection	Rules 1 and 2
DMZ Services	Rules 3 and 4
Outbound Access	Rules 5, 6 and 7
Cleanup	Rule 8

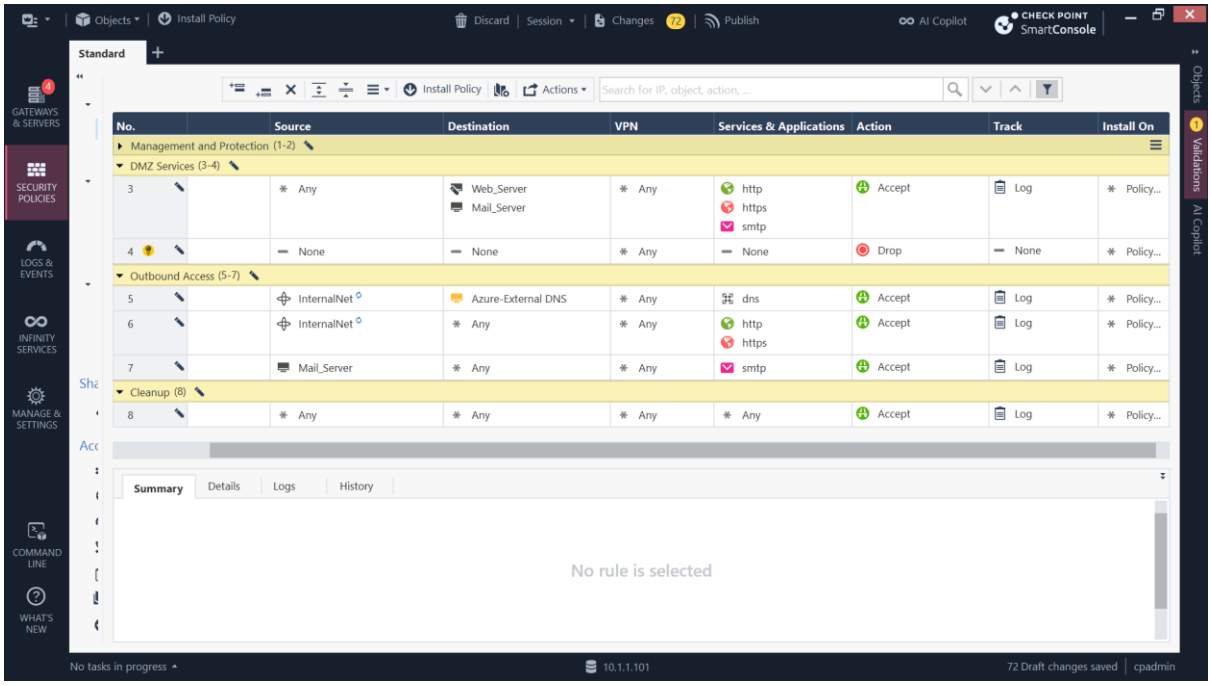
Procedure

1. Opened the Access Control rulebase in SmartConsole.
2. Right-clicked the rulebase and selected New Section Title.
3. Created the four required sections.
4. Moved each rule into its appropriate section.
5. Collapsed and expanded a section to verify functionality.
6. Edited a section title and then restored the original name.



Rulebase with all sections expanded showing:

- Management and Protection
- DMZ Services
- Outbound Access
- Cleanup



Rulebase with one section collapsed showing:

- Successful collapse/expand functionality
- Organized rule grouping

Result

The rulebase was successfully divided into four logical sections. All rules were placed in their correct categories, and section management features were verified

Observation

Organizing rules into sections improves readability and makes large rulebases easier to manage. Administrators can quickly locate related rules without affecting the logic of the security policy.

Do Sections Affect Rule Enforcement?

No. Sections are used only for organization and administration. The Security Gateway receives and enforces the rules in their existing top-to-bottom order, regardless of the section titles. Sections help administrators manage the policy but do not change how traffic is evaluated.

Conclusion

The Access Control rulebase was successfully organized into four sections. The sections improved clarity and policy management while maintaining the original rule processing order and enforcement behavior.

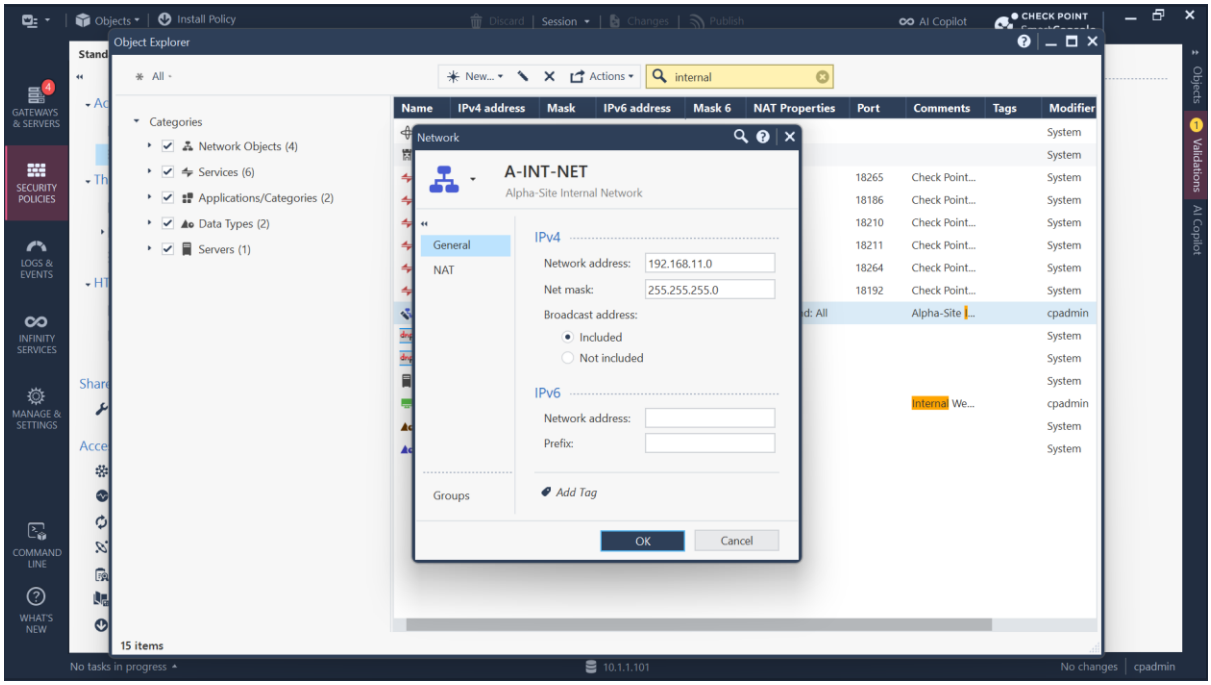
Task 6: Configure Hide NAT

Objective

To configure automatic Hide NAT for the Internal_LAN network, examine the NAT rule generated by SmartConsole, and understand how many-to-one address translation works.

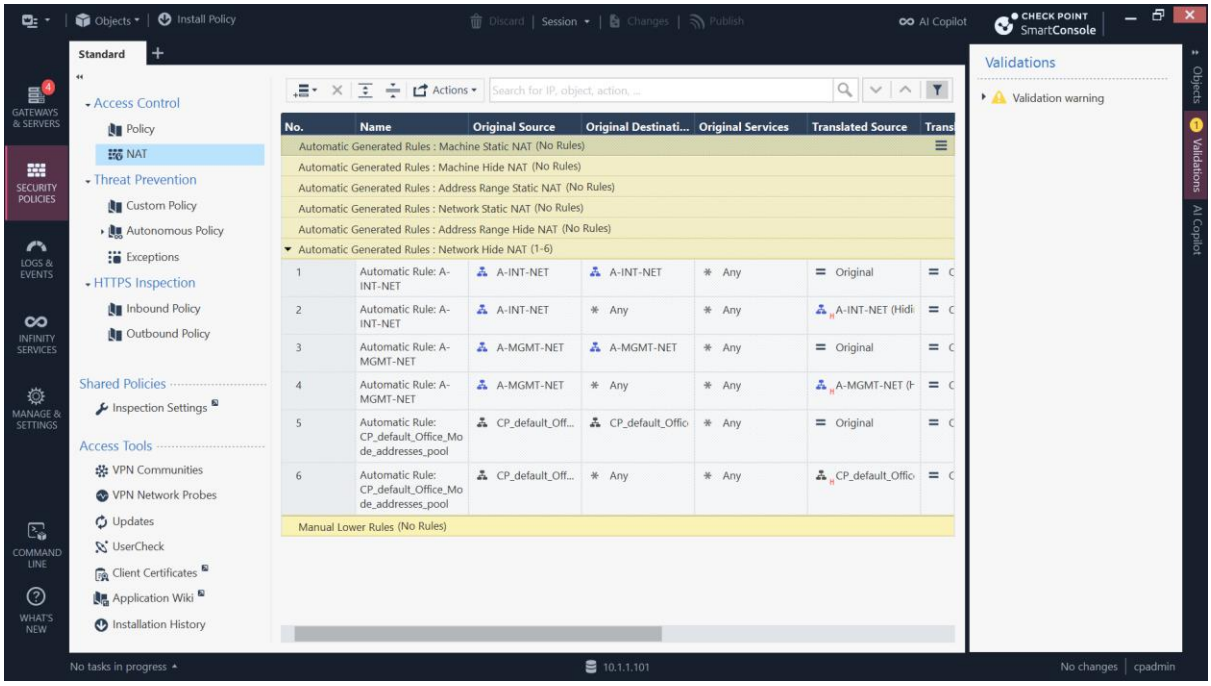
Procedure

1. Opened the Internal_LAN object in SmartConsole.
2. Navigated to the NAT tab.
3. Enabled Add automatic address translation rules.
4. Selected Hide as the translation method.
5. Chose Hide network behind Gateway.
6. Saved the configuration and published the changes.
7. Opened Security Policies → NAT and located the automatically generated NAT rule



Internal_LAN NAT settings showing:

- Automatic NAT enabled
- Hide NAT selected
- Hide behind Gateway configured



NAT rulebase displaying:

- Automatically generated Internal_LAN NAT rule
- Source translation settings
- NAT rule created by SmartConsole

Result

Hide NAT was successfully configured for the Internal_LAN object. SmartConsole automatically generated the required NAT rule in the NAT rulebase.

Why This Is an Automatic NAT Rule

This is an automatic NAT rule because it was created automatically by SmartConsole after NAT settings were enabled on the Internal_LAN object. The administrator defined the NAT behavior through the object's properties rather than manually creating a separate NAT rule in the NAT rulebase.

What Many-to-One Translation Means

Many-to-one translation means that multiple internal devices share a single external IP address when accessing external networks. Instead of each device using its private address, all outbound traffic appears to originate from the gateway's public address.

What Happens to the Source Port

When multiple devices use the same translated IP address, the gateway modifies and tracks source port numbers as required. This process allows the gateway to distinguish between different client connections and correctly return responses to the originating device.

Observation
Hide NAT improves security by concealing internal IP addresses from external networks. It also conserves public IP addresses because many internal hosts can share a single translated address through the gateway.

Conclusion
The Internal_LAN network was successfully configured with automatic Hide NAT. SmartConsole automatically generated the corresponding NAT rule, allowing multiple internal hosts to share the gateway's translated address while maintaining separate connections through source port translation.

Task 7: Rule Order Experiment and Installation History

Objective
To demonstrate how rule order affects traffic matching, verify policy installation results, and review deployment records using Installation History**.**

Procedure

Phase 1: Rule Above Internal Web Out

1. Created a temporary rule named TEMP Block Internal.
2. Placed the rule above Internal Web Out.
3. Published the changes.
4. Installed the policy.
5. Repeated test T4 and observed the matching rule.

Phase 2: Rule Below Internal Web Out

1. Moved TEMP Block Internal below Internal Web Out.
2. Published the changes.
3. Installed the policy again.
4. Repeated test T4 and observed the matching rule.

Phase 3: Cleanup

1. Deleted the temporary rule.
2. Published the changes.
3. Installed the policy.

4. Opened Installation History and reviewed the installation containing the temporary rule.

Why Specific Rules Belong Above General Rules

Specific rules should be placed above general rules because Check Point evaluates rules from top to bottom. If a broad rule is placed first, it may match traffic before a more specific rule has a chance to be evaluated. Placing specific rules higher in the rulebase ensures that security policies are enforced as intended and reduces the risk of unintended access.

Observation

The experiment demonstrated that rule order directly affects policy enforcement. Although the same rules existed in both tests, changing the rule position changed the final action taken on the traffic.

Conclusion

The experiment confirmed that Check Point Security Gateways evaluate rules sequentially from top to bottom. The first matching rule is enforced, making rule placement a critical part of policy design and security management**.**

Part B: Theory Questions

Section B1: Multiple choice

Q1. Where is a Security Policy created, stored and enforced?
ANS: B

Q2. Which control model does the Check Point Access Control firewall policy use?
ANS: A

Q3. Where should the Stealth rule be placed?
ANS: C

Q4. Which statement about rulebase sections is correct?
ANS: D

Q5. What type of translation does Hide NAT perform?
ANS: B

Section B2: Short answers

Q6. Distinguish between Explicit rules and Implied rules. State where Implied rules are configured and where they can be placed. What is the Implicit Cleanup Action?
ANS: Explicit rules are rules that administrators manually create in the rulebase, while Implied rules are automatically applied by Check Point for management and communication purposes. Implied rules are configured in Global Properties and can be placed before, after, or within the rulebase. The Implicit Cleanup Action is the action applied to traffic that does not match any rule, usually Drop.

Q7. Explain why rule order matters in a rulebase. Give one example from your own rulebase.
ANS: Rule order matters because Check Point evaluates rules from top to bottom and applies the first matching rule.

Q8. Explain the difference between saving and publishing changes in SmartConsole. What happens to unpublished changes when you install a policy, and why do many organisations publish only during a change window?
ANS: Saving stores changes only in the current administrator session, while Publishing commits the changes to the Management Server database and makes them visible to other administrators. Unpublished changes are not included when a policy is installed. Many organizations publish only during approved change windows to reduce operational risk and maintain change control

Q9. Compare Global Properties with Policy Packages. Give one example of what each controls and say how you open each in SmartConsole.
ANS: Global Properties contain settings that affect the overall Check Point environment, such All devices Managed by the Security Management. You can open them through Application Menu → Global Properties/ Manage Policies and Layers. Policy Packages contain Access Control While Global properties Contains NAT.

Q10. Name the four predefined Security Zones. State two rules about assigning gateway interfaces to zones, and one benefit of using zone-based rules.
ANS: The four predefined Security Zones are Internal, External, DMZ, and Wireless. A gateway interface can belong to only one zone, and interfaces should be assigned to the zone that matches their network purpose. The main benefit of zone-based rules is that they simplify policy management by allowing rules to be written for zones instead of individual interfaces.

Section B3: Scenario questions

Q11. A new administrator inserts a rule with Source Any, Destination Any, Services Any and Action Accept above the Stealth rule. Describe two security problems this causes,

explain how to correct it, and name the type of rule that is allowed to sit above the Stealth rule and why.

ANS: Placing an Accept Any rule above the Stealth Rule creates two problems: it allows unrestricted access to the network and bypasses security controls by matching traffic before other rules are checked. To correct this, remove the overly permissive rule or place it below the required security rules. The only rule that should normally be above the Stealth Rule is an administrative access rule (SSH/HTTPS management access), so administrators can manage the gateway.

Q12. After Hide NAT is configured for Internal_LAN, staff report they cannot reach the Internet.

(a). Two Benefits of NAT

1. NAT hides internal IP addresses from external networks, improving security.
2. NAT allows many internal devices to share a single public IP address, conserving public address space.

(b). Three Checks to Find the Fault

1. Check the NAT rulebase to confirm that the automatic Hide NAT rule for Internal_LAN was created correctly.
2. Check the Access Control policy to verify that rules such as DNS Out and Internal Web Out are allowing the required traffic.
3. Check Logs & Events to determine whether traffic is being accepted, dropped, or failing due to another policy or configuration issue.